LESSON 01 / 03

The focal properties are geometric angle laws

A parabola sends rays parallel to its axis through the focus. At an ellipse point, the tangent makes equal angles with the two focal segments, so a ray from one focus reflects toward the other. For a hyperbola, a directed ray toward one focus reflects so that its line passes through the other focus; state the directed-ray convention carefully because a hyperbola has two branches.

A convenient unifying statement uses the normal: at a central conic, the normal makes equal angles with the lines joining the point to the two foci. For the parabola, the normal bisects the angle between the focal segment and a line parallel to the axis.

SHORT DERIVATION

Why the ellipse reflects between its foci

Along a line $PF_1 + PF_2$ changes by the sum of two projected displacements. On a tangent to the level curve $PF_1 + PF_2 = 2a$, the total first-order change is zero. Hence the two unit focal directions have opposite tangential projections, which is exactly the equal-angle law. The same level-set idea gives the directed hyperbola version.

Confocal means shared foci, not merely shared centre

An ellipse and hyperbola are confocal when their foci coincide. In centred horizontal form,

$$\frac{x^2}{A} + \frac{y^2}{B} = 1, \qquad A > B > 0,$$

has $c^2 = A - B$, while

$$\frac{x^2}{C} - \frac{y^2}{D} = 1$$

has $c^2 = C + D$. Confocality therefore requires $A - B = C + D$. Two confocal central conics through a common point have orthogonal tangents there.

WORKED EXAMPLE

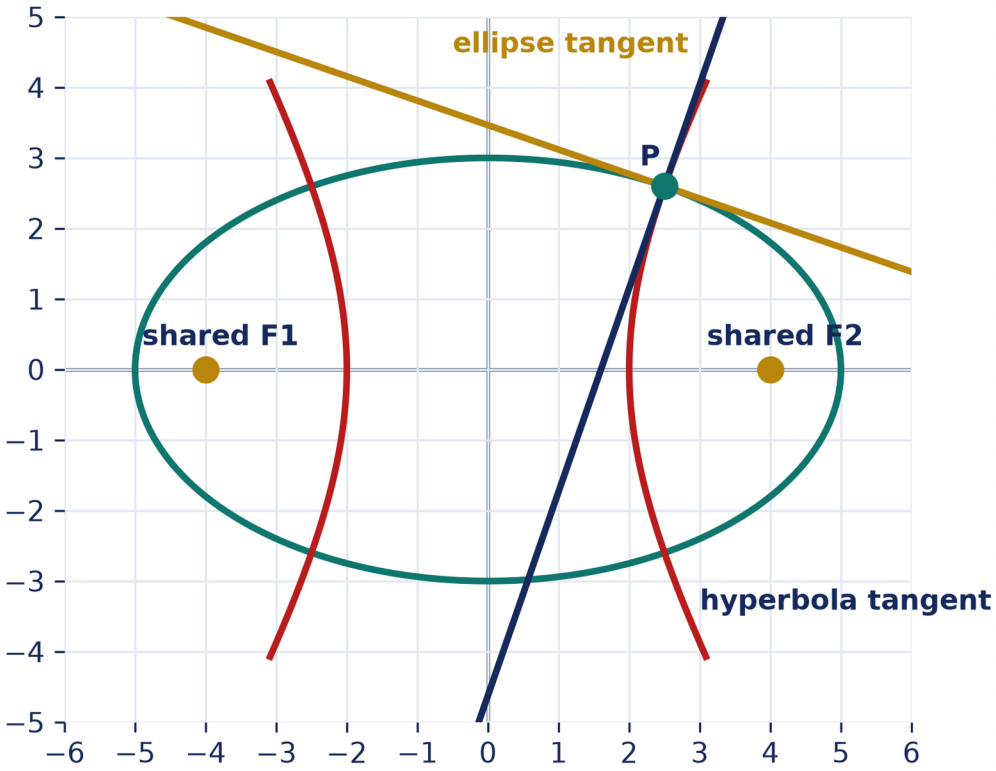
An exact orthogonal confocal intersection

The ellipse $x^2/25 + y^2/9 = 1$ and hyperbola $x^2/4 - y^2/12 = 1$ both have foci $(\pm 4, 0)$ because $25 - 9 = 4 + 12 = 16$. They meet at

$$P = \left(\frac{5}{2}, \frac{3\sqrt{3}}{2}\right).$$

The tangent slopes at P are respectively $-\sqrt{3}/5$ and $5/\sqrt{3}$, whose product is -1. Thus the two curves and their tangent lines are orthogonal at P.

Confocal ellipse and hyperbola meet orthogonally



READ THE FIGURE

The ellipse is x^2/25 + y^2/9 = 1 and the hyperbola is x^2/4 - y^2/12 = 1. Both use c^2 = 16. The tangent slopes are -sqrt(3)/5 and 5/sqrt(3), whose product is -1.

KEY RESULT

Two useful problem transformations

Reflection problems often become straight-line problems after reflecting a focus across a tangent. Confocal problems often become elliptic-coordinate problems after introducing a parameter λ and studying $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$. Check the denominator signs to determine whether a member is an ellipse, hyperbola, or degenerate locus.

COMMON TRAP

Do not reverse the reflection direction

An ellipse ray from one focus reaches the other. A hyperbola statement needs directed lines and branch awareness; saying simply that both foci always receive the same physical ray is imprecise. Draw the incoming and reflected directions before applying an optical argument.

LESSON 02 / 03

Four concyclic-style incidences on $xy = c^2$

Take three noncollinear points on a rectangular hyperbola, with distinct nonzero parameters:

$$A = (ct_1, \tfrac{c}{t_1}), \quad B = (ct_2, \tfrac{c}{t_2}), \quad C = (ct_3, \tfrac{c}{t_3}).$$

The slope of AB is $-1/(t_1t_2)$, so an altitude perpendicular to it has slope t_1t_2. Imposing the two altitude conditions gives the orthocentre

$$H = (-\tfrac{c}{t_1t_2t_3}, -ct_1t_2t_3).$$

This point also lies on $xy = c^2$, because its coordinates have product c^2. Thus, for a nondegenerate triangle inscribed in a rectangular hyperbola, **the orthocentre lies on the same hyperbola**.

SHORT DERIVATION

Why the formula works

The altitude through A is perpendicular to BC, whose slope is $-1/(t_2t_3)$; its slope is therefore t_2t_3. The line through H and A has exactly that slope after the parameter formula is substituted. Cyclic permutation verifies the other two altitudes, so all three concur at H.

WORKED EXAMPLE

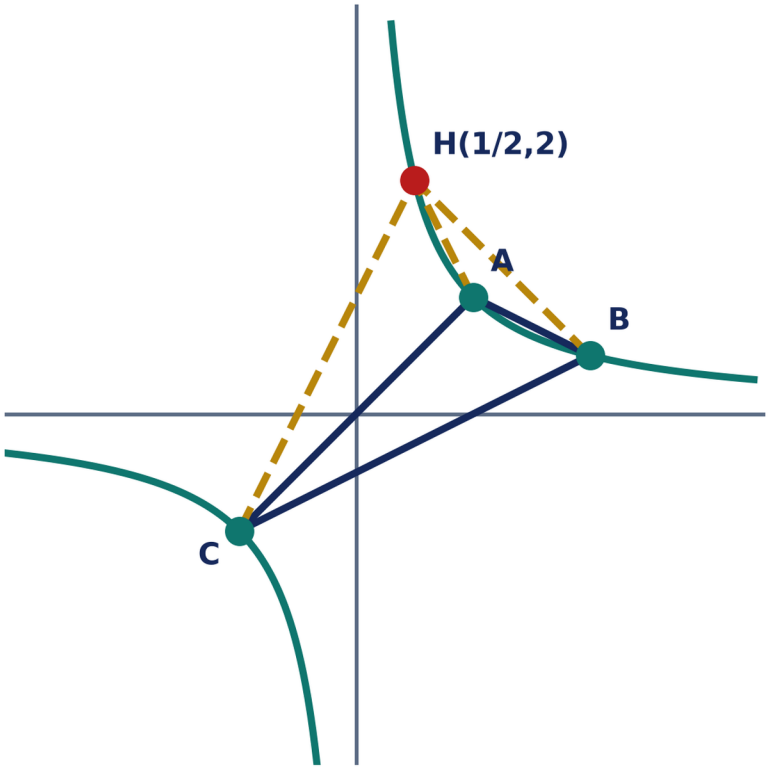
A small exact configuration

On $xy = 1$, take parameters $1, 2, -1$:

$$A = (1, 1), \qquad B = (2, \tfrac{1}{2}), \qquad C = (-1, -1).$$

Their product is -2, so the formula gives $H = (1/2, 2)$. Indeed BC has slope $1/2$, and AH has slope -2; AC has slope 1, and BH has slope -1. All four points lie on $xy = 1$.

A, B, C and their orthocentre H all lie on xy = 1



READ THE FIGURE
The dashed segments lie along the three altitudes. The slope checks AH perpendicular BC and BH perpendicular AC are exact; the third altitude follows by concurrence.

Connect the configuration to the Euler line

For any triangle with circumcentre O, orthocentre H, and nine-point centre N,

$$N = \frac{O+H}{2}, \qquad R_{\text{nine-point}} = \frac{R_{\text{circum}}}{2}.$$

These are general triangle facts, not special hypotheses about the hyperbola. The rectangular-hyperbola theorem adds the powerful incidence $H \in xy = c^2$; combining it with Euler-line geometry often exposes a midpoint, circle, or reflection that coordinates alone would hide.

KEY RESULT
A productive proof pattern
Parameterize three points by t_1, t_2, t_3, compute a slope as $-1/(t_i t_j)$, and use the orthocentre formula when perpendicularity appears. If the problem involves a fourth point on the same rectangular hyperbola, test whether it is the orthocentre of the other three before introducing more variables.

LESSON 03 / 03

Read a general conic through invariants

Write

$$F(x,y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$

and define

$$I = a + b, \qquad \delta = ab - h^2, \qquad \Delta = a(bc - f^2) - h(hc - fg) + g(hf - bg).$$

Here Δ is the determinant of the symmetric coefficient matrix with rows (a,h,g), (h,b,f) and (g,f,c). The quadratic part controls directions at infinity, while Δ detects many degeneracies. For nondegenerate real conics: $\delta > 0$ is elliptic, $\delta = 0$ parabolic, and $\delta < 0$ hyperbolic. In the elliptic case, $\Delta \neq 0$ and $I\Delta < 0$ gives a real ellipse; $I\Delta > 0$ gives no real locus.

COMMON TRAP

Use signs and the coefficient convention together

For example, $x^2 + y^2 + 1 = 0$ has $\delta > 0$ but no real points. When $\Delta = 0$, factor or complete squares to distinguish points, line pairs and empty loci. Preserve the coefficients $2h, 2g, 2f$; nonzero scaling leaves the locus unchanged but scales Δ, so use nonvanishing and consistent sign patterns, never the determinant's numerical size.

General conic decision table for real coefficients

Algebraic signal	Nondegenerate reading	Degenerate caution
delta > 0, Delta != 0	elliptic; use I*Delta for reality	Delta = 0 can collapse to a point
delta = 0, Delta != 0	parabola	Delta = 0 can give parallel/repeated lines or no r
delta < 0, Delta != 0	hyperbola	Delta = 0 can give two intersecting lines
Delta = 0	do not use the nondegenerate rule	factor or complete squares to identify the actua

READ THE FIGURE

These are classification guides, not substitutes for checking real points and degeneracy. The notation Delta is the 3 by 3 determinant and delta = ab - h^2.

Rotation removes the cross term

A rotation of axes through an angle θ satisfying

$$\tan(2\theta) = \frac{2h}{a-b}$$

removes the xy term; when $a = b \neq 0$, use $\theta = \pi/4$. Translations then complete the squares for central conics. Orthogonal changes preserve lengths and angles, so they are preferable when the geometry itself matters.

WORKED EXAMPLE

Classify and rotate in one example

For

$$F = x^2 + 4xy + y^2 - 1 = 0,$$

we have $a = b = 1, h = 2, c = -1$, so $\delta = 1 - 4 = -3$ and $\Delta = 3 \neq 0$. It is a hyperbola. Set $x = (X - Y)/\sqrt{2}$ and $y = (X + Y)/\sqrt{2}$; then

$$F = 3X^2 - Y^2 - 1.$$

Thus the rotated standard form is $3X^2 - Y^2 = 1$, whose asymptotes are $Y = \pm\sqrt{3}X$ in the rotated coordinates.

Degeneracy is exact algebra, not a drawing judgement

If the full expression factors into real linear factors, the locus is the union of their lines. For instance,

$$4x^2 - 12xy + 9y^2 = (2x - 3y)^2 = 0$$

is one repeated line, while $x^2 - y^2 = (x - y)(x + y)$ is two intersecting lines. Both have $\Delta = 0$. Completing squares can also expose a point or an empty real locus.

Pascal and Brianchon: projective companions

For a nondegenerate conic, **Pascal's theorem** says that for a hexagon inscribed in the conic, the intersections of the three pairs of opposite sides are collinear. **Brianchon's theorem** is dual: for a hexagon whose sides are tangent to a conic, the three principal diagonals are concurrent. Degenerate conic versions give classical line theorems such as Pappus.

KEY RESULT

When these tools help

Use invariants for classification without heavy rotation, and Pascal/Brianchon for incidence problems with many intersections or tangents. They are projective statements: keep track of the six-point or six-line ordering, and allow parallel-line intersections to be interpreted at infinity when needed.